Characterizing Gamma-Ray Spectrograms of Cs-137, Na-22, and Co-60

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The spectra of Cs-137, Co-60, and Na-22 have been explored using an NaI scintillator and pulse-height analyzer (PHA). A tantalum-backed source of Cs-137 was observed to have $\sim 50\%$ higher count rate in the backscatter regime (200 keV) than when backed with lead. Possibilities of a combined lead-tantalum response due to thin backing (< 2 cm) are discussed. This Cs-137 source was then used to calibrate the PHA, and energy peaks for Co-60 ($\gamma_1 = 1208.0 \pm 0.4$ keV, $\gamma_2 = 1383.8 \pm 0.5$ keV) and Na-22 ($\gamma = 1318.5 \pm 0.5$ keV, $Annihilation = 522.7 \pm 0.2$ keV) were obtained. Gaussian + Quadratic curves were used for peak fitting.

I. INTRODUCTION

Using photomultiplier tubes and scintillators to characterize radiation sources is a delicate task, since gamma ray spectrograms are especially sensitive to calibration and their surrounding environment. Sources must be surrounded with some sort of shielding (usually lead) to protect the user, and will interact with any nearby media. Scintillators convert radiation into photons that can be captured by a photomultiplier, but may not respond to all energy levels proportionally. Analog calibration of the photomultiplier's signal, as well as digital calibration of produced spectra, can also shift and warp features of these spectra significantly.

Monte-Carlo simulations of radiation sources can predict certain features of their spectra, but are limited by how accurate their modeling is. Programs like P-TreCK can keep track of Compton scattering, photoelectric absorption, and pair production interactions within a scintillator, but struggle to account for more complex interactions with surrounding shielding and internal x-ray production [1]. Thus, collecting physical data is important for understanding what radiation is received by a real detector, especially in the lower-energy regime.

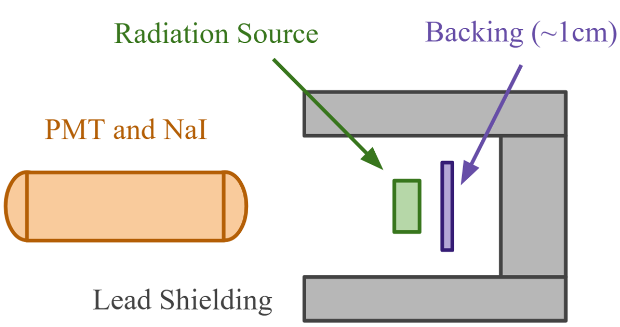

FIG. 1. A diagram of the experimental setup. The radiation sources used include Cs-137, Co-60, and Na-22. Sources were either backed immediately with a thin layer of lead (Pb) or tantalum (Ta) of ∼1 cm thickness and surrounded by thick lead blocks on all sides. The scintillator used is a cylinder of NaI (Ti) with a diameter of 1.5" and a length of 1.75".

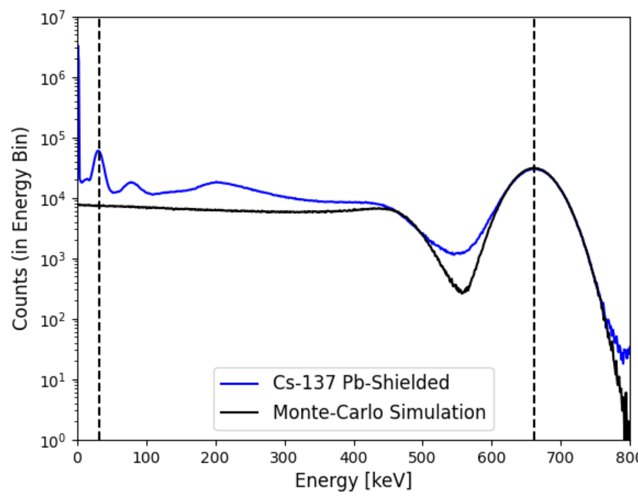

FIG. 2. The spectrogram of Cs-137 (in blue) compared to the Monte-Carlo Model. The black dotted lines at 31 keV and 661 keV show the peaks used to calibrate the pulse-height analyzer. The model accounts for the main gamma ray peak, Compton Scattering, x-ray photo-absorption, and any secondary effects, but does not account for the lead shielding. Each bin is approximately 1.5 keV wide.

When detectors are well-calibrated to known radiation sources, they can be used to measure the spectra of other sources and validate their expected decay chains. This relies on accurately fitting peaks to particular decay values, which can require complicated fit equations if several features in spectrum overlap [2]. Keeping track of overall uncertainties is more difficult for this reason.

The main features of spectra produced are the results of the photoelectric absorption, Compton scattering, and pair production. In photoelectric absorption, the energy of a gamma particle or photon ejects an inner-shell electron of a material. When another electron then fills this vacancy, a photon is emitted. In the cases where the original incident particle does not reach the detector, a sharp peak forms in the spectrum associated with this energy emission known as an x-ray fluorescence peak. At lower energy levels (below 100 keV), photoelectric absorption is a dominant effect.

Compton scattering occurs when an incident particle

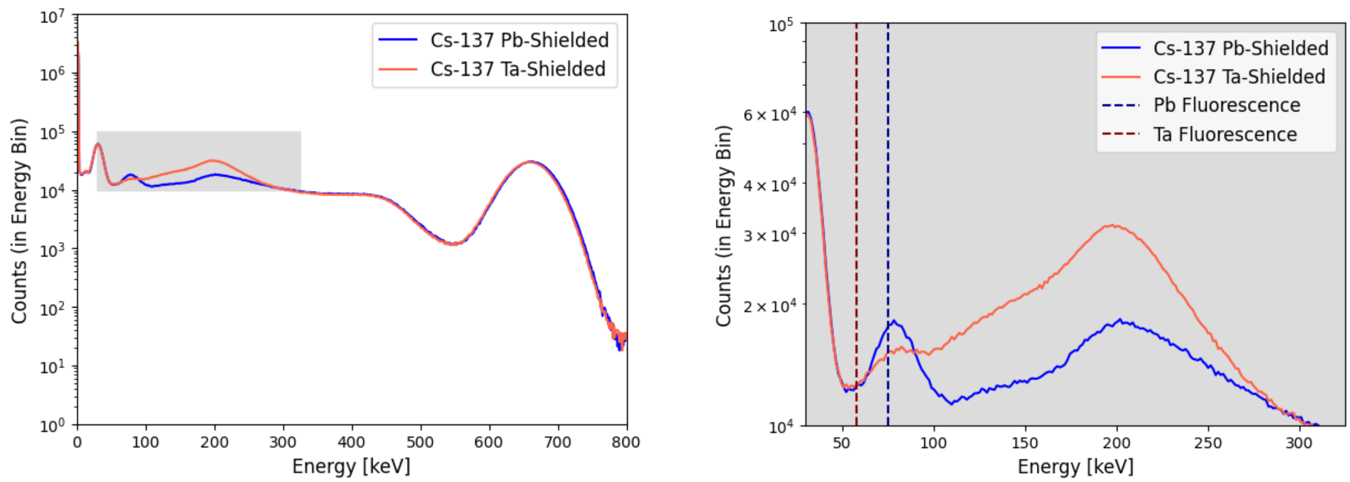

FIG. 3. The spectrogram of Cs-137 when backed with lead (blue) and tantalum (orange). The full spectrogram is shown on the left, and a zoomed-in portion of the spectrogram is shown on the right. Note the region between 60 keV and 270 keV where the spectra diverge. The photo-absorption peaks for both Lead and Tantalum are drawn with dotted purple and red lines, respectively.

collides with an outer-shell electron, resulting in two outbound particles. Compton scattering can happen at a variety of angles, producing a continuous spectrum, but has a minimum possible energy transfer at 180°. Thus, a characteristic Compton shelf forms in spectra at

$$E_{Compton\,Shelf} = E_\gamma - \frac{E_\gamma}{1 + 2\frac{E_\gamma}{m_e c^2}} \tag{1}$$

since Compton-scattered particles cannot have a final energy between their initial gamma energy and this value.

Compton scattering can also occur off of shielding materials such as lead. This then generates a backscatter peak for particles that lose a maximum amount of energy when they are reflected, at

$$E_{Backscatter\,Peak} = \frac{E_\gamma}{1 + 2\frac{E_\gamma}{m_e c^2}} \tag{2}$$

The backscatter and Compton shelf energies sum to the original incident gamma energy.

Lastly, at high enough energies, gamma radiation can collide with a nucleus and generate an electron-positron pair, known as pair production. For this to occur, the incident energy needs to be able to create both particles, so the threshold for pair production occurring is:

$$E_{Pair\,Production} = 2m_e c^2 = 1022\;keV \tag{3}$$

The positron eventually will be annihilated, which will produce 511 keV photons and a subsequent peak there in a spectrum.

II. EXPERIMENT AND CALIBRATION

Fig. 1 shows a schematic of the experiment. Radiation sources of Cs-137, Co-60, and Na-22 were placed in a thick lead housing, and backed with a thin layer of either lead or tantalum. An NaI (Ti) scintillator and photomultiplier tube (PMT) assembly was positioned roughly 4” from the radiation source, powered using a high voltage amplifier. This assembly was then fed into an Ortec Pulse-Height Analyzer (PHA) to capture the spectra of the samples. Data for the Cs-137 and Na-22 samples was collected for 30 minutes, and data for the Co-60 sample was collected for 2 hours. The amplifier was adjusted following the guidelines from the user manual and Confluence experiment page.

The spectra were then analyzed using the Interspec software package. In addition to the analog calibration, the software used reference data for Cs-137 and applied a linear fit using the 661 keV gamma ray peak and 32.2 keV x-ray peak, shown in Fig. 2 with dotted black lines. This linear calibration was then used for analyzing the spectra for Co-60 and Na-22.

Fig. 2 also plots a Monte-Carlo simulation of the Cs-137 source produced using the P-TreCK software [1]. This simulation captures primary and secondary events of Compton scattering and photoelectric absorption, but does not model any shielding or background effects outside of the source and the detector. The simulation has a relatively flat count rate between 0 and 400 keV where x-ray and backscatter peaks appear to occur in the observed Cs-137 spectrum, but the simulation matches the main gamma peak and Compton shelf quite well.

III. RESULTS

Fig. 3 shows overlapping spectra of Cs-137 taken with different shielding (either lead or tantalum) behind the source. The plot on the left shows that both spectra are very nearly aligned at high energies (∼270 keV and up) as well as very low energies (∼60 keV and below), but diverge significantly at energies between those bounds.

The plot on the right is a zoomed-in portion of the spectra to show more detail. At around 200 keV, both graphs appear to have peaks, but the tantalum-shielded spectrum has around 50% more detector counts. Moving down in energy, the tantalum spectrum linearly decreases before it reaches a plateau between 70-100 keV, while the lead spectrum has a more pronounced peak around 70 keV. Reference x-ray fluorescence peaks are shown for lead (in dotted blue) and tantalum (in dotted red), at 75.0 keV and 57.5 keV, respectively. Both spectra appear to have some response near the lead x-ray energy and neither have a response near the tantalum x-ray energy.

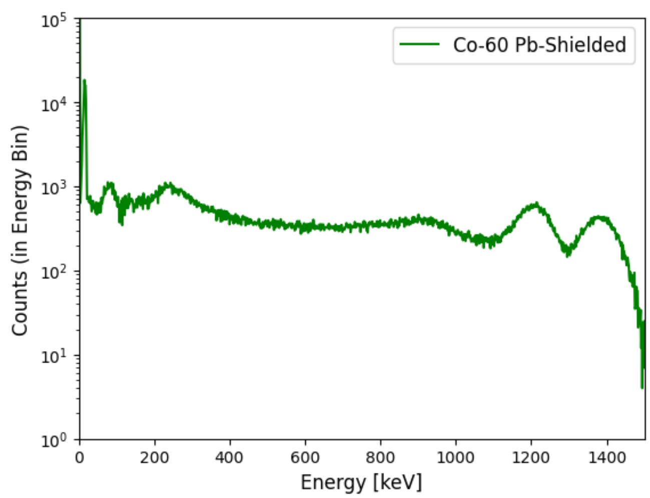

FIG. 4. The spectrogram of Co-60 in green. The two gamma ray peaks are located at 1136.7 ± 0.4 keV and 1301.9 ± 0.4 keV. These peaks were calibrated using the Cs-137 spectrum and fit using a Quadratic + Gaussian method in the Interspec software package.

Using the calibration from the Cs-137 spectrum, spectra for both Co-60 and Na-22 were produced which can be seen in Fig. 4 and Fig. 5, respectively. For the Co-60, there appear to be two distinct peaks at higher energies, which were fit using Interspec's software package. A Gaussian fit was applied over an underlying quadratic fit of the spectrum, taking the overlap of nearby peaks into consideration. Using this method, the peaks were determined to be centered at $\gamma_1 = 1136.7 \pm 0.4$ keV and $\gamma_2 = 1301.9 \pm 0.4$ keV, with a 1-σ uncertainty from a χ^2 fit.

The rest of the Co-60 spectrum appears quite flat, with a very faint shelf around 1 MeV. Peaks similar to those in the lead-backed Cs-137 spectrum are found around 250 keV and 80 keV, with a third sharp peak down around 15 keV.

The same peak-fitting methods were also applied to the Na-22 spectrum found in Fig. 5. The lower-energy peak is fitted to $\gamma_1 = 522.7 \pm 0.2$ keV and the later $\gamma_2 = 1318.5 \pm 0.5$ keV. Compared to the previous spectra, there is a noticeable difference in the count rate from 0 keV leading up to the first peak (around 350 keV) compared to the count rate from 600 keV to 1000 keV between the two peaks. Both peaks also have small shelves on their lower-energy side.

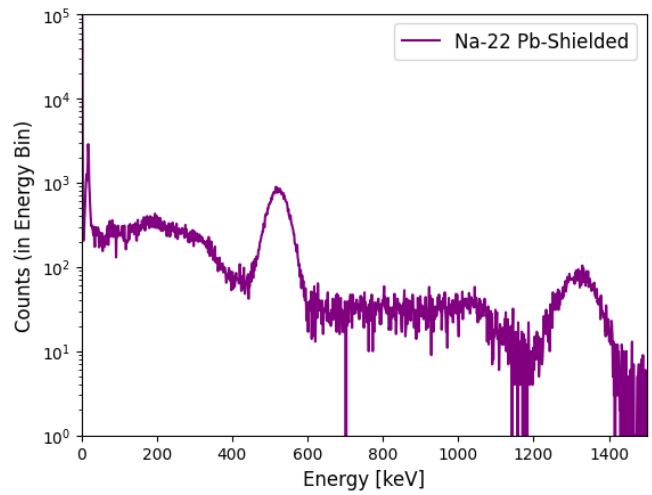

FIG. 5. The spectrogram of Na-22 is shown in purple. A gamma ray peaks is located at 1318.5 ± 0.5 keV, with another significant peak at 522.7 ± 0.2 keV.

From Fig. 3, it is clear that the surrounding shielding of the source is having a substantial impact on the Cs-137 spectra, namely in the x-ray regime and around 200 keV. The backscatter equation predicts a peak at 184.32 keV for Cs-137, which appears to be reasonably represented by the data. Since backscatter peaks come from ∼180° Compton scattering, they are skewed to the right since angles just under 180° are possible but angles greater than that are not.

Interestingly, the amount of backscatter is greater for the tantalum-backed source rather than the lead-backed source, which is counter-intuitive when considering lead's atomic number (82) is greater than that of tantalum (73). At first glance, one would expect a more positively-charged nucleus to repel incoming electrons more effectively, creating a higher backscatter signal. However, since the tantalum-backing was relatively thin and still surrounded by a large amount of lead backing, it is likely that the tantalum spectrogram is subject to a more composite effect.

Previous work has established that it is possible for materials with a lower atomic number to exhibit more backscatter, such as comparing wood to steel wool [5]. In these cases, the targets with higher atomic numbers can absorb and remove more photons via their own pho-

TABLE I. A summary of the measured peaks for Co-60 and Na-22 compared to reference values.

Energy Peak	Reference	Measured	Absolute Difference	Percent Difference
Co-60 γ_1	1175 ± 6 [3]	1208.0 ± 0.4	$+33 \pm 6$	$+2.8 \pm 0.5\%$
Co-60 γ_2	1331.8 ± 1.0 [3]	1383.8 ± 0.5	$+52.0 \pm 1.5$	$+3.83 \pm 0.08\%$
Na-22 γ	1277 ± 4 [3]	1318.5 ± 0.5	$+41.5 \pm 4$	$+3.2 \pm 0.3\%$
Na-22 Annihilation	511 ± 4 [4]	522.7 ± 0.2	$+11.7 \pm 4$	$+2.3 \pm 0.8\%$

toelectric effect. In Fig. 3, we see that the lead-backed spectrum does have more counts around the known x-ray fluorescence peak for lead than the tantalum-backed spectrum, so this could be an indication that this phenomenon is indeed decreasing the backscatter reaching the detector. The tantalum also appears to generating little-to-no x-ray fluorescence at its known peak.

More recent analysis of the saturation depth of a variety of materials would indicate that high-atomic number backings with thicknesses below 1.5 cm may not be fully saturated [6]. That is, the tantalum-backing is thin enough where scattering events that happen beyond it (such as in the lead housing behind it) can reflect back towards the detector without complete absorption by the tantalum. This could explain the smaller but still noticeable response of the tantalum-shielded spectrum around the lead x-ray fluorescence peak, and suggest physically how the combined backscatter from the tantalum and the lead behind it could be greater than just the lead alone. The lack of tantalum fluorescence may also be evidence of the low attenuation required to make this possible.

For both the Co-60 in Fig. 4 and the Na-22 in Fig. 5, the measured peak energies are significantly greater than other reference values, which is potentially indicative of systematic error [3, 7]. These results are summarized in Table III.

Calibrating the PHA from just two peaks in the Cs-137 spectrum (the main gamma peak and the smaller x-ray peak) means that only a linear fit could be used (in addition to the analog calibration). NaI scintillators are known to have a non-linear response, tending to react more to lower-energy gamma rays than higher-energy gamma rays [8]. Since the PHA was calibrated using 32 keV and 661 keV peaks, it is possible that it would "overshoot" at the higher-energy peaks of Co-60 and Na-22, which could explain why the gamma peaks are consistently greater than their corresponding reference values.

This non-linearity argument, however, does not account for the overshoot in the Na-22 annihilation peak, which sits below the Cs-137 661 keV peak. It is possible that this difference could be due to using the x-ray peak, since NaI detectors have a large drop-off in resolution below 100 keV [9]. This low-resolution, combined with the contribution of many other possible Cs-137 x-rays with lower count rates, could account for the overall shift in the Na-22 annihilation peak and the other gamma peaks as a whole.

Other gamma-ray detecting techniques outside of NaI scintillation could be used to resolve the resolution issues described above. Alternate scintillators, such as lanthanum bromide ($LaBr_3$), have been shown to have significantly better resolution compared to NaI and are more effective at capturing higher-energy radiation [10]. HPGe detectors require cooling to operate, but are also especially good at sorting tightly-packed x-ray peaks [11]. At much higher (tens of MeV) and much lower (below 30 keV) energies, pair spectrometers and gas ionization detectors respectively are more effective measurement devices. Of these options, moving to a $LaBr_3$ scintillator could be the easiest way to both improve the x-ray peak calibration of the Cs-137 and reduce non-linear effects.

IV. CONCLUSION

The effect of the tantalum backing of the Cs-137 source is surprising both for the increased backscatter relative to the lead backing and the limited fluorescence response of the tantalum. The thinness of the tantalum is a likely culprit of this result, but further testing would be needed to confirm if compounding scattering and attenuation effects between the tantalum and lead could cause such an increase in backscatter. A more intense Monte-Carlo simulation, such as Geant4, may also be able to provide additional insight into this result [12].

The energies found for the gamma and annihilation peaks of Co-60 and Na-22 do not agree particularly well with other established references, but the limited calibration resources and more basic uncertainty propagation could have a heavy impact on the results found from this data. The use of additional radiation sources to calibrate the PHA could significantly increase the accuracy of the data collected, and a more thorough treatment of systematic uncertainties (possibly also studied more through simulations) could give a better overall estimate of error.

Energy Peak	Photoelectric	Compton	Pair Prod.
Cs-137 γ	11.6%	88.4%	0.0%
Co-60 γ_1	4.6%	95.2%	0.2%
Co-60 γ_2	3.9%	95.2%	0.9%
Na-22 γ	4.1%	95.3%	0.6%

TABLE A. Relative cross-sections of the gamma ray energy peaks, using data from NIST [13].

Appendix A: Supplementary Information

NIST reports the mass attenuation coefficient for 1 MeV gamma rays in NaI as $\mu/\rho = 0.0588\ g/cm^2$ and the density as $\rho = 3.67\ g/cm^3$ [13]. Thus, the mean free path can be calculated as

$$\lambda = \frac{1}{\mu} = \frac{1}{\mu/\rho \times \rho} = 4.63\ cm. \tag{A1}$$

So, the probability that a particular 1 MeV gamma ray does not interact with the 1.75" long scintillator can be calculated as

$$P_{No\ Interaction} = e^{-\mu L} = 38.3\% \tag{A2}$$

Of course, particles that do interact within the scintillator may have multiple interactions, and not all of these generated photons will reach the PMT.

NIST also reports the relative mass attenuation coefficients for the photoelectric effect, Compton scattering, and pair production [13]. The ratio of the cross-sections are summarized in table A.

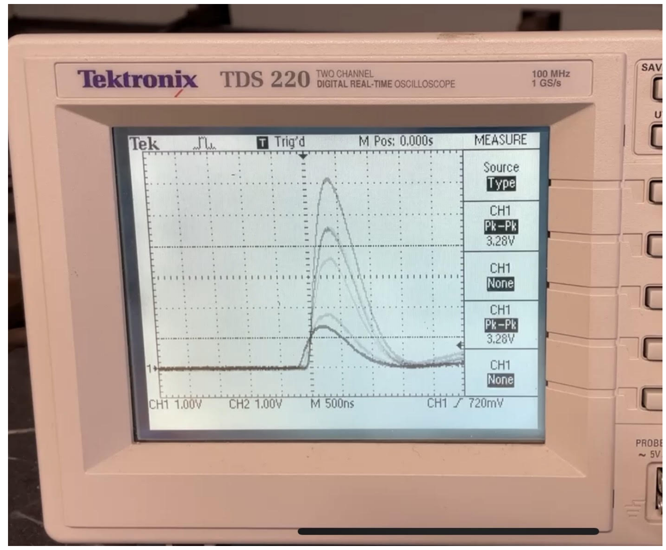

FIG. 6. A picture of the oscilloscope and sample trace used to help physically calibrate the PMT and amplifier assembly. The average pulse height was about 3V (dotted horizontal line), but pulse heights greatly varied.

When physically calibrating the apparatus, an oscilloscope was used to help ensure that pulses entering the PHA were reasonably large (varied in voltage) while avoiding saturating larger pulses. The amplifier connected to the PMT was tuned so that the average pulse height in the distribution was around 3V, as shown in Fig 6. The window and trigger of the scope was also adjusted to confirm that large pulses were largely unaffected.

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